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An ISO 9001:2015 Certified Organisation

Sixth Weekly Advisory for Cotton Cultivation from 16th to 22nd July, 2026

A. General advisory on soil health, nutrition, pest, and disease management

Higher cotton yields rely on balanced soil nutrition, water management and proactive, monitoring-based pest and disease management. Implementing a holistic and integrated crop management plan ensures proper growth and vigour, reduces input costs, and protects the crop from devastating pest and disease damage.

Soil health & nutrition management (INM)

- **Optimize macronutrients**: Improve soil health by prioritizing crop rotation, organic amendments, and cover crops, and by using rational chemical fertilization based on pre-sowing soil testing.
- **Apply micronutrients:** Address secondary and micronutrient deficiencies (such as Zinc, Boron, Manganese, and Magnesium) proactively via targeted foliar sprays as needed.

Disease management (IDM)

- **Deploy preventative practices**: Management of both soil-borne and foliar pathogens should be achieved through integrated cultural practices, including proper field sanitation and the application of biopesticides and biofertilizers as seed and soil treatments to enhance plant health and suppress disease incidence.
- **Control foliar outbreaks:** Regular field monitoring should be undertaken during periods of high humidity and recommended fungicides should be applied judiciously only when prevailing environmental conditions are conducive to rapid disease development and spread.

Integrated pest management (IPM)

- **Conduct field scouting:** Perform weekly field scouting to monitor sucking pests (such as thrips, aphids, jassids and spider mites) alongside bollworms.
- **Utilize traps & bio-pesticides:** Deploy pheromone traps early in the season to track adult moth flights, predict egg-laying cycles and apply organic, natural or plant-based pesticides as a primary response.
- **Conserve natural enemies:** Safeguard beneficial insects like lady beetles, lacewings and big-eyed bugs by choosing highly selective insecticides over broad-spectrum formulas.
- **Restrict chemical interventions:** Use label-recommended chemical pesticides strictly as a last resort, deploying them only when pest populations reach established economic thresholds.

B. Location-specific advisory

PUNJAB		Actual Rainfall in last week(mm)							Predicted Rainfall in next week (mm)				
		July							July				
		07	08	09	10	11	12	13	15	16	17	18	19
	Firozpur								2.2	1.2	0.3	0.2	0.1
	Faridkot	0	0	22.8	0	0	1.2	0	3.2	0.6	0.1	0.3	0.2
	Muktsar								3.3	0.2	0.1	0.3	0.2
	Bhatinda	0	1.8	0	0	0	0	0	3	0.6	0.2	0.4	0.5
	Sangrur								1.8	0.6	0.2	0.8	0.9
	Ludhiana	0	0	0	0	42	0	0	2.1	1	0.1	0.8	0.6
Amount of rainfall & colour Code		0.1 to 2.4 mm		2.5 to 15.5 mm		15.6 to 64.4 mm				64.5 to 115.5 mm		115.6 to 204.4	
Rainfall category		Very light rainfall		Light rainfall		Moderate rainfall				Heavy rainfall		Very heavy rainfall	

Crop condition

At Bathinda, the crop is 50 to 70 days old at the flowering stage. Hoeing, manual weeding and intercultural operations are in progress. Applied the second split of Nitrogen fertilizer. Weeds like *Chulai (Amaranthus viridus), Dodhak* (*Euphorbia* spp)*, Tandla (Digera arvensis), Madhana (Eleusine spp), Trianthema monogyna,* Makru (Ipomea spp), Khabbal (*Cynodon dactylon*), Chibber bel (*Cucumis trigonus*) have infested the fields. Need-based sprays based on the ETL of each pest were given to the affected fields. No incidence of diseases.

At Faridkot, the crop is 58 to 68 days at square formation and flowering stage in a timely-sown crop. Intercultural operations were carried out. Manual or tractor operated interculture operations are done to control emerging weeds. At a few isolated places, weeds like *Trianthema* spp. (Itsit), *Digera* spp. (Tandla) have emerged after the application of irrigation or rainfall, but are controlled as per recommended herbicides. Manual hand hoeing has been done to remove leftover weeds from intra-row spaces. Whitefly population was near ETL at almost all the locations except two locations crossing ETL. Thrips population was very high (above ETL) at all locations with clear cut silvery appearance and erect leaves throughout the fields. The incidence of TSV was also spotted at a few locations with low severity. Pink bollworm incidence on flowers was low at most of the locations, however, high trap catches were observed in the AI traps installed in Muktsar Sahib. DCB incidence was noticed in all the fields in squares and flowers. In some fields, square drying was also noticed as well as some spots of Muktsar sahib showed stressed flowering due to lack of irrigation. The PBW Pheromone traps were also distributed to the farmers in contact.

Advisory:

At Bathinda, the overall crop stand is good. In case of heavy rains, farmers are advised to drain out the excessive water from cotton fields. Remove weeds in and around the fields. Use low-cost yellow sticky traps @ 40 per ha during the initial crop phase to check early infestation of whitefly in the field. Do not grow *bhendi*, *moong* and *arhar* in the cotton crop and as border rows to minimize the infestations of spotted and American bollworms, and sucking pests (jassid and whitefly). Apply whitefly management treatments when the population reaches 6 adults/leaf in the upper canopy or honeydew appear on 50% of the plants. For jassid, initiate spaying as soon as 50% of the plants display marginal yellowing and curling on mature upper leaves. For thrips, apply spray treatment when nymphal and adult populations reach 12/leaf in the upper canopy of plants. Spray the crop as soon as the crawlers/adults of mealybugs appear on the cotton plant. At 60 days after sowing, spray Profenophos 50% EC @ 500 ml per acre to control thrips and PBW, and Flonicamid 50%WG @ 80g per acre using 200 litres of water for controlling whitefly. If the whitefly population ranges between 4-6 adults per leaf in cotton fields, then spray neem-based insecticides @ 1 litre per acre or neem seed kernel extract (NSKE) @ 1200 ml per acre (use 200 litres of water per acre for spraying). Install pheromone traps @ 2 per acre for monitoring of PBW moths. For early-sown crops (first fortnight of April), apply the first spray against pink bollworm when 10-20 per cent plants start producing squares, particularly in the fields adjoining cotton ginning mills, to prevent initial build-up of the pest population. Apply subsequent sprays at 5% damage in flowers and green bolls (ETL >5%). Remove 'rosette flowers', if any and destroy them. Wherever the crop is at the flowering stage, spray Emamectin benzoate 5 SG @ 100 g/acre or Spinetoram 11.7 SC @ 170 ml/acre using 200 litres of water, if green boll infestation is above 5%. After the first irrigation, if incidence of parawilt is observed, spray a cobalt chloride solution @1 g/100 litres of water immediately after the appearance of symptoms on the affected plants and repeat twice at 5-7 day intervals.

At Faridkot, farmers are advised to drain out excessive water in the event of heavy rainfall as cotton is very sensitive to stagnating water. After heavy rains or irrigation, some plants might show wilting due to Para wilt which can be checked by spraying Cobalt chloride @ 10 mg/ liter of water on the affected plants at the initial stage of wilt. Give first irrigation 4-6 weeks after sowing, depending upon soil type, which also helps in managing thrips infestation. Apply 120 kg Urea per acre in 2-3 equal splits, depending upon soil type and moisture conditions (Avoid Urea broadcasting just before irrigation,). Spray 1,250 ml per hectare of Pyrithiobac sodium 6% + Quizalofop ethyl 4% (a post-emergence weedicide) in 500 litres of water to control annual grass and broadleaf weeds after the first irrigation. Alternatively, manual or tractor-operated interculture operations should be done to control emerging weeds. In case of high thrips infestation, apply light irrigation for its management. If the infestation increases, spray Profenofos 50 EC @ 500 ml/acre (in 200 litres of water for one acre). Monitor pink bollworm using funnel or delta traps baited with Gossyplure and installed at about 15 cm above the crop canopy. Replace the lure after every 15 days and use 5 traps/ha. Keep the fields and fallow/barren areas around the farm/canals, etc., free from weeds , which act as collateral hosts of leaf curl virus. For management of thrips-transmitted Tobacco Streak Virus (TSV), strictly follow field sanitation, particularly the removal of *Parthenium* weed in and around the cotton field, and recommended management practices for thrips.

HARYANA		Actual Rainfall in last week(mm)							Predicted Rainfall in next week (mm)				
		July							July				
		07	08	09	10	11	12	13	15	16	17	18	19
	Hisar	0	0	0	0	14.8	0	0	1.2	0.2	0.3	0.2	0.9
	Jind								1.8	0.5	0.2	0.5	2.6
	Sirsa								1.8	0.4	0.4	0.2	0.2
	Rohtak								1	0.3	0.4	0.3	2.2
Amount of rainfall & colour Code		0.1 to 2.4 mm		2.5 to 15.5 mm		15.6 to 64.4 mm			64.5 to 115.5 mm			115.6 to 204.4	
Rainfall category		Very light rainfall		Light rainfall		Moderate rainfall			Heavy rainfall			Very heavy rainfall	

Crop Condition:

At Hisar, the crop is at the squaring stage. Weeding and hand hoeing were taken up. Weeds like *motha, santhi* and *makra* have spread in the fields. Initial populations of thrips, whitefly, and jassid were noticed but well below the economic threshold.

At Sirsa, the crop is 48 to 78 days old at the vegetative growth and flowering stages. Hand hoeing, manual weeding, and other intercultural operations by tractor-mounted cultivator or bullock and camel are in progress. Weed appearance was noticed at a few locations. Stressed flowering is also observed at a few locations. The population of sucking pests recorded below ETL and in traces at all surveyed locations. The population of whitefly & jassids was recorded below ETL at all locations surveyed and the thrips population was above ETL at one location & population ranged between 09-37 nymphs & adults per 3 leaves. Spotted bollworm in Desi cotton was recorded below ETL.

Advisory:

At Hisar, farmers are advised to apply one bag of Urea after rains or the first irrigation in a more than seven-week-old crop and do hand hoeing at an appropriate stage once rains get over. The population of sucking pests is below ETL, and therefore, there is no need to spray chemical insecticides. Install gossyplure baited pheromone traps @ 2/acre to monitor pink bollworm moths in more than 45 days old crop and record the population of moths at 3-day intervals. Neem-based insecticides may be sprayed @ 1 litre per acre (use 200 litres of water for one acre) if moth catches exceed ET (5-8 moths per trap per night for 3 consecutive days). Destroy the dried cotton stalks of the previous crop season, if any are left in the field. Examine at least 100 flowers per acre for pink bollworm infestation and remove and destroy the damaged flowers (rosette) along with the larvae. In root rot-affected patches, drench the affected plants with Carbendazim 50 WP @ 10 g/ 10 litres of water or *Trichoderma viride* or *T. harzianum* @ 100 g/ 10 litres of water.

At Sirsa, farmers are advised to irrigate the field if the crop has attained an age of more than or equal to 35 days to avoid heat stress. Apply the first split dose of urea fertiliser after irrigation or rain if the crop attains an age of more than or equal to 35 days. Take up hand hoeing with *kasola / khurpa* or intercultural operations with a tractor/bullock/camel-operated *triphali* to keep the crop weed-free. Field and field bunds should be weed-free, and *Parthenium* weed should be eradicated in and around the fields to minimize the CLCuD and TSV incidence. Alternatively, spray 1,250 ml per hectare Pyrithiobac sodium 6% + Quizalofop ethyl 4% to control annual grass and broadleaf weeds or spray 1,250 ml per hectare or 2,250 ml per hectare Glufosinate ammonium as a directed spray (using a protective hood) to control weeds in between the crop rows. Use 200 litres of water per acre for spraying. Install yellow sticky traps @ 8-10 per acre to monitor whitefly population. Install

pheromone traps (2/acre) for pink bollworm moth activity, monitor the rosette flowers by random sampling of 100 flowers/acre at regular intervals at 35 days after sowing to monitor PBW infestation. Keep the field, field bunds and surrounding free from weeds, cucurbits and other plants that act as alternate hosts for whitefly and reservoirs of cotton leaf curl virus. For management of sucking pests (whitefly, thrips), apply neem oil or NSKE 5 % @ 1.0 litre per acre using 200 litres of water up to 60 days after sowing (DAS) and if crop age more than 60 days then apply emamectin benzoate 5SG @ 100 g or spinetoram 11.7 SC @ 170 ml or profenophos 50 EC @ 500 ml per acre (use 200 litres water per acre for spraying of insecticides). These insecticides are also effective against pink bollworm. For spotted bollworm management in Desi (*arboreum*) cotton, apply a prophylactic spray of cypermethrin10 EC @ 200-300 ml **or** cypermethrin 25 EC @ 80-100 ml **or** deltamethrin 2.8 EC@ 160-200 ml **or** fenvalerate 20 EC@ 100-200 ml **or** fenpropathrin10 EC @ 300 ml per acre if infestation exceeds 10%. Keep monitoring root rot and wilt disease incidence regularly following the first irrigation. Under heavy rainfall and waterlogging conditions, maintain proper drainage in the field to remove excess water from the field to prevent crop damage. Drenching early symptomatic plants and surrounding plants with *Trichoderma* spp. (*T. harzianum or T. viride*) 1% WP @ 100 g o r Carbendazim 50%WP @ 10 g/ 10 L of water to manage root-rot and wilt disease.

RAJASTHAN		Actual Rainfall in last week(mm)							Predicted Rainfall in next week (mm)				
		July							July				
		07	08	09	10	11	12	13	15	16	17	18	19
	Ajmer	0	0	0	0	8.7	0	1.8	0	0	0	0	1.8
	Jodhpur	20.3	0	0	0	0	0	0	0	0	0	0	0
	Nagaur								0	0	0	0	0.2
	Pali								0	0.1	0.2	0	0
	Sri Ganganagar	0	0	0	0	0	0	0	1	0	0.1	0	0.1
Amount of rainfall & colour Code		0.1 to 2.4 mm		2.5 to 15.5 mm		15.6 to 64.4 mm			64.5 to 115.5 mm			115.6 to 204.4	
Rainfall category		Very light rainfall		Light rainfall		Moderate rainfall			Heavy rainfall			Very heavy rainfall	

Crop Condition:

In Sriganganagar and Hanumangarh districts, the crop is 54 to 70 days old at the vegetative and flowering stages. Post-sowing, irrigation has been applied at some places; manual hoeing/weeding and intercultural operations are going on. Weeds like Itsit (*Trianthema spp.*), tandla (*Digera arvensis*) and Motha (*Cyperus rotundus*) have infested the crop. Whitefly incidence noticed from 2.18 to 5.70/3 leaves, thrips population 3.60 to 12.23/3 leaves and jassids from 0 to 0.60/3 leaves.

At Banswara, the crop is 15 to 21 days old at the germination stage. Farmers with irrigation facilities have started sowing cotton. In the rest of the areas, fields are ready to take up sowing cotton. No incidence of pests or diseases.

Advisory:

In Sriganganagar and Hanumangarh districts, farmers are advised to apply the recommended split dose of nitrogenous fertilisers after the first irrigation for maximum fertiliser use efficiency. Avoid broadcast application of nitrogenous fertilizers before irrigation, as this leads to leaching of fertilizers. Remove weeds in and around the cotton fields. Eradicate *Parthenium* weeds around the fields to minimize TSV incidence. Monitor the crop for insect pests and diseases regularly. Install yellow sticky traps @ 8-10 per acre to monitor whitefly population. Install pheromone traps @ 5/ha (2/acre) to monitor pink bollworm moth activity. Spray neem-based insecticides @ 1 litre per acre if moth catches exceed ETL (5-8 moths per trap per night for 3 consecutive days). Remove and destroy the damaged flowers (rosette) along with larvae.

In southern Rajasthan (Banswara, Bhilwara, Chittorgargh, Dungarpur, Pratapgarh, Rajsamand, Salumber, and Udaipur), the crop will be sown after the onset of monsoon. Farmers are advised to arrange fertilizers, FYM, and seeds for sowing of crops well in advance. Select sucking pest and disease-tolerant, and early-maturing varieties/hybrids of cotton, which fit in cotton-wheat rotation. Those farmers having irrigation facilities should start sowing cotton with a spacing 90 × 45 cm. Apply a basal dose of fertilisers (40:60:30 NPK kg/ha at the time of sowing cotton. Before sowing, treat seeds with Carboxin 37.5% + Thiram 37.5% DS @ 3.5 g per kg of seeds (root rot and bacterial diseases) or Fluxapyroxad (333 g/L FS) @ 1.5 ml per kg seed or Tetraconazole 11.6% W/W (12.5% w/v) SL @ 1. 5 ml per kg of seeds to manage seed-borne diseases (for root rot disease).

ODISHA		Actual Rainfall in last week(mm)							Predicted Rainfall in next week (mm)				
		July							July				
		07	08	09	10	11	12	13	15	16	17	18	19
	Koraput	0.5	0	0.4	1.5	0	0	0	3	3	9.4	14.6	7.3
	Kalahandi	0	1.2	0	0.6	6.8	0	0	3.5	14	23	30.6	14.9
	Balangir								6	15	29	25.6	17.7
Amount of rainfall & colour Code		0.1 to 2.4 mm		2.5 to 15.5 mm		15.6 to 64.4 mm			64.5 to 115.5 mm			115.6 to 204.4	
Rainfall category		Very light rainfall		Light rainfall		Moderate rainfall			Heavy rainfall			Very heavy rainfall	

Crop Condition:

In Odisha, the crop is 15 to19 days old at the seedling and early vegetative stage. Land preparation, application of FYM and basal dose of fertilizers, sowing of cotton, intercrops, border crops, and green manuring crops, application of pre-emergence herbicides, gap filling were the operations taken up during the reporting period. All types of weeds i.e, broad leaf, grasses and sedges were noticed at the seedling stage but the weed population is low due to the pre-emergence application of pendimethalin. No incidence of pests and diseases was observed.

Advisory:

Farmers are advised to complete sowing of cotton by 10th July to avoid late-season drought situations. Take up sowing as soon as monsoon rain starts. Carry out gap filling in sown fields at 7 DAS to maintain the required plant population. Take up intercultural operations and weeding, drainage of excess rainwater from the field. Give the first top dressing of fertilizer at 25-30 DAS with 50% N and 50% K. Provide drainage channels in the fields to avoid waterlogging that will hamper germination of seeds and growth of seedlings. In case of moist soil conditions where manual weeding is not possible, go for the application of herbicides like Quizalofop ethyl 5 % EC @ 2 ml per litre of water (for grassy weeds) and Pyrithiobac sodium 10% EC @ 1.5 ml per litre of water (for broad-leaf weeds) at 20-25 DAS. Regularly monitor the crop to know about the incidence of any pests and diseases. To prevent sucking pests like aphids, spray neem-based pesticide @ 1 litre/ acre. Use Pendimethalin 30 EC @ 1.33 litres in 200 litres of water for one acre as a pre-emergence spray at 1-2 days after sowing to control weeds in the recently sown crop.

GUJARAT		Actual Rainfall in last week(mm)							Predicted Rainfall in next week (mm)				
		July							July				
		07	08	09	10	11	12	13	15	16	17	18	19
	Amreli								0.5	0.7	0.7	0.8	1.1
	Bhavnagar								0.2	0.4	0.4	0.5	0.6
	Jamnagar	3	0	0	0	0	0	0	0.5	0.7	0.7	1	1.2
	Rajkot	5.6	0	0	0	0	0	0	0.5	0.8	0.9	1.2	1.4
	Junagadh	7.2	6.6	0	0	0	0	0	1.2	1.7	1.6	2	2.9
	Sabarkantha								0.3	1.2	0.9	0.8	1.3
	Surendranagar								0.2	0.2	0.8	0.6	0.5
	Ahmedabad	4.4	13.2	0	0	0	0	0	0.1	0.3	0.6	0.4	0.8
	Baroda								0.4	0.7	1.1	2.1	2.2
	Patan								0.3	0.5	1.1	0.7	1.2
	Mehesana								0.2	0.8	0.6	0.5	1
Amount of rainfall & colour Code		0.1 to 2.4 mm		2.5 to 15.5 mm		15.6 to 64.4 mm			64.5 to 115.5 mm			115.6 to 204.4	
Rainfall category		Very light rainfall		Light rainfall		Moderate rainfall			Heavy rainfall			Very heavy rainfall	

Crop Condition:

In Surat, the sown crop is in the germination stage. Formation of furrows and the cleaning of fields is taken up. In some pockets, sowing is in progress where rainfall was sufficient. Removed excess rain water from the field where excess rainfall was received.

At Junagadh, sufficient rainfall has been received at many places during the last week. However, in some places, rainfall is awaited till date. Sowing is completed at few and under progress in many places. The sown crop is at germination and early vegetative stage

Advisory:

At Surat, farmers are advised to carry out field sanitation. Those farmers having sufficient irrigation facilities can start sowing cotton in their fields. Use Pendimethalin 38.7 % CS @ 700 ml/acre as pre-emergence weedicide to control weeds in the early stage of the crop. Take up sowing of cotton crop only after receipt of 75 -100 mm of monsoon rainfall. In case of heavy rain, drain out the excess rainwater from the cotton fields. Cotton crops require 240-40-00 NPK/ ha. Dose of Nitrogen is applied after sowing @ 30, 60, 75, 90 and 105 days in equal splits. Apply a complete dose of Phosphorus at the time of sowing as a basal dose. Give post-emergence spray of Quizalofop ethyl @ 2 ml per litre of water to 15-day crop with one hand weeding as the most effective method to manage weeds. Check *Fusarium* wilt patches along with a one-meter radius of healthy plants in desi cotton fields with suitable fungicide application.

At Junagadh, farmers are advised to purchase recommended Bt varieties/Bt hybrid seeds with the original bill. Those farmers having sufficient irrigation facilities are advised to start sowing cotton. Give a basal dose of chemical fertilizers viz; Phosphorus 25 kg/ha and Potassium 75 kg/ha (Nitrogen missing) to irrigated sown crop. To manage soil-borne diseases, at the time of sowing, treat the seeds with Carboxin 37.5% + Thiram 37.5% DS) @3.5 g per kg of seeds or *Trichoderma harzanium* 2.5 kg/ha enriched with 100 kg FYM for field application. Those farmers having sufficient irrigation facilities are advised to start the pre-sowing of cotton. Use micro irrigation system to save water

MADHYA PRADESH		Actual Rainfall in last week(mm)							Predicted Rainfall in next week (mm)				
		July							July				
		07	08	09	10	11	12	13	15	16	17	18	19
	Khargaon								0.1	0.3	0.1	0.3	0.9
	Dhar								0.4	0.4	0.4	0.4	1
	Khandwa								0	0.1	0.1	0.4	2
Amount of rainfall & colour Code		0.1 to 2.4 mm		2.5 to 15.5 mm		15.6 to 64.4 mm			64.5 to 115.5 mm		115.6 to 204.4		
Rainfall category		Very light rainfall		Light rainfall		Moderate rainfall			Heavy rainfall		Very heavy rainfall		

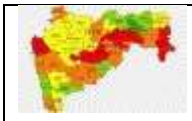
Crop Condition:

At Khandwa, the cotton crop is 7 to 52 days old at the initial stage and the vegetative growth stage based on the period sown. On an average more than 50% of cotton area has already sown where the irrigation facility was available. Rains were received on alternate days during the reporting period. Incidence of jassids, aphid and whiteflies has been observed in some fields. Summer season weeds like *Cynodon dactylon, Cyperus rotundus, Argemone maxicana* and *Phylanthus niruri* have infested the fields

Advisory

Farmers are advised to apply the first split dose of Urea fertiliser after irrigation or rain if the crop attains an age of more than or equal to 35 days and it should be applied by column method at the depth of 10 to 15 cm. Take up hand hoeing or intercultural operations with a tractor/bullock-operated implement to keep the crop weed-free. Field and field bunds should be weed-free, and *Parthenium* weed should be eradicated in and around the fields to minimize the TSV and sucking pest incidence. Under heavy rainfall and waterlogging conditions, maintain proper drainage in the field to remove excess water from the field to prevent crop damage. Apply drenching of copper oxychloride 50 % WP @ 250 g **or** Carbendazim 50% WP @ 120 g plus Urea 2 kg per 100 litres of water twice at one-week intervals starting immediately after symptom development. For management of sucking pests- whitefly, thrips- apply neem oil or NSKE 5 % @ 1.0 litre per acre using 200 litres of water up to 60 days after sowing (DAS).

MAHARASHTRA		Actual Rainfall in last week(mm)							Predicted Rainfall in next week (mm)				
		July							July				
		07	08	09	10	11	12	13	15	16	17	18	19
	Dhule								0.2	0.4	0.4	0.5	0.6
	Nandurbar								0.8	1.2	0.8	1.2	1.8
	Jalgaon	0	21	35	0	0	0	0	0	0.2	0.2	0.5	0.7
	Ahilyanagar	16.2	23	20	0	0	0	0	0.2	0.8	0.7	0.6	1
	Ch. Sambhaji Nagar								0	0	0.2	0.6	1.7
	Jalna								0	0	0.2	0.6	1.7
	Beed								0	0.1	0.2	0.7	1.2
	Nanded								0	0.1	0.2	5.8	1.5



	Parbhani	1.4	0	2.8	0	0	0	0	0	0.1	0.1	2.6	2.9
	Hingoli								0	0.1	0.2	2.2	1.9
	Buldhana								0	0	0.1	0.2	1.7
	Akola	0	13.3						0	0	0.1	0.2	1.1
	Washim								0	0	0.1	0.8	1.2
	Amravati								0.1	0.2	0.1	1	2.8
	Yavatmal								0	0	0.4	2.1	1.6
	Wardha								0	0.1	0.3	2.5	5.9
	Nagpur								0	0.3	0.7	8.3	7.8
	Chandrapur	3	1	3.4	8.8	0	0	0	0	0	2	6.6	5.4
Amount of rainfall & colour Code		0.1 to 2.4 mm		2.5 to 15.5 mm		15.6 to 64.4 mm				64.5 to 115.5 mm		115.6 to 204.4	
Rainfall category		Very light rainfall		Light rainfall		Moderate rainfall				Heavy rainfall		Very heavy rainfall	

Crop Condition:

At Akola, sowing of cotton is in progress. Delayed due to the late onset of the monsoon. Land preparation for the coming *kharif* season is completed. All the operations like harrowing, levelling, FYM application, etc. are completed. Field layout is in progress for sowing cotton. No incidence of pests and diseases.

At Rahuri, the sown irrigated cotton crop is 8 to 43 days at germination, seedling to initial vegetative stages based on the time sown. Sowing is completed in the majority of the fields. Gap filling, irrigation, weeding and fertilizer application are in progress. Weeds like *Parthenium, Haryali, Lavala, Dudhi, Kena, Nagarnotha, chandvel,* etc., have infested the fields. No incidence of pests and diseases.

At Nanded, the sown crop is 11 to 36 days at initial vegetative stages. Gap filling and thinning have been taken up. Land preparation, harrowing, levelling, FYM application, etc. have been completed by rainfed farmers. Sowing is to be initiated once sufficient rainfall is received in rainfed farms. No incidence of pests and diseases.

Advisory:

At Akola, farmers are advised to take up sowing cotton after receipt of sufficient rainfall of 75-100 mm. Undertake gap filling in cotton wherever gaps are observed in the sown fields. Avoid cultivation of *Bt* cotton on light types of soil. Use short duration *Bt* /non *Bt* varieties/hybrids for rainfed sowing and mid-late to late varieties/hybrids for irrigated conditions. If the commercially available cotton seeds are not pre-treated with fungicides, treat the cotton seeds with Carboxin 37.5% + Thiram 37.5% DS @ 3.5 g per kg of seeds (root rot and bacterial blight disease) or Fluxapyroxad (333g/lit FS) @ 1.5 ml/kg seed or Tetraconazole 11.6% w/w (12.5% w/v) SL @ 1.5 ml/kg of seeds for seed- borne diseases (for root rot disease), before sowing. Also give seed treatment of *Azotobacter* for nitrogen fixation and PSB @20-25 gm /kg seed for phosphate solubilization. Include intercrops like green gram and black gram in cotton in 1:1 row proportion. Use pendimethalin 38.7 % CS @ 700 ml/acre in 200 litres of water as a pre-emergence weedicide to control weeds in the early stage of crop. Do not use pre-emergence herbicides in dry conditions. Follow recommended spacing and fertilizer doses for *arboreum* (60x15 cm, 60x30cm, 40:20:20 Kg NPK/ha), Improved *hirsutum* (60x30 cm, 60:30:30 Kg NPK/ha) and rainfed Bt hybrid cotton (90x45, 90x60, 60:30:30 Kg NPK/ha) and irrigated *Bt* Cotton (120x30,120x60 cm,120:60:60 Kg NPK/ha, respectively

At Rahuri, farmers are advised to avoid early sowing of the crop. Start sowing after receipt of sufficient rainfall of 75-100 mm. Those farmers having sufficient irrigation facilities can start sowing cotton in their fields. Use short duration *Bt* / non *Bt* varieties/hybrids for rainfed sowing and mid-late to late varieties/hybrids for irrigated conditions. In case the commercially available cotton seeds are not pre-treated with fungicides, farmers are advised to practice seed treatment for management of seed borne diseases with Carboxin 37.5% + Thiram 37.5% DS @3.5 g per kg of seeds (root rot and bacterial blight disease) or Fluxapyroxad (333g/lit FS) @ 1.5 ml/kg seed or Tetraconazole 11.6% w/w (12.5% w/v) SL @1.5 ml/kg of seeds (for root rot disease). Also give seed treatment of *Azotobacter* for Nitrogen fixation and PSB @20-25 gm /kg seed for phosphate solubilization. Include intercrops like green gram and black gram in cotton in a 1:1 row proportion. Take up sowing of cowpea, maize, *Setaria,* and marigold as trap crops. Field and field bunds should be weed-free, and Parthenium weed should be eradicated in and around the fields to minimize the thrips population build-up and TSV incidence.

At Nanded, farmers are advised to take up sowing once sufficient soil moisture is available. Take up sowing of cotton 120 x 45 cm for rainfed conditions. In case of High-Density Planting System, adopt spacing at 90 x 15 cm under rainfed conditions on medium / red soils, whereas closer spacing at 90 x 30 cm in heavy, deep soils and under irrigated conditions. For gap filling

treat the seed with fungicide Carboxin 37.5% + Thiram 37.5% DS @ 3.5 g per kg of seeds (root rot and bacterial blight disease) or Fluxapyroxad (333g/lit FS) @ 1.5 ml/kg@1.5ml/kg seed or Tetraconazole 11.6% w/w (12.5% w/v) SL @ 1.5 ml/kg of seeds for seed-borne borne diseases (for root rot disease). as well as biofertilizers, *viz.,* Azotobacter, PSB and KMB @ 10 ml per kg seed. Give a basal dose of NPK @60:75:75 kg/ha to irrigated cotton if not applied at sowing time and a basal dose of 48:60:60 NPK kg/ha to the rainfed crop before or at the time of sowing cotton. Use pendimethalin 38.7 % CS @ 700 ml/acre in 200 litres of water as a pre-emergence herbicide to control weeds in the early stage of the crop. Field and field bunds should be kept weed-free, and *Parthenium* weed should be eradicated in and around the fields to minimize the thrips population build-up and TSV incidence.

Contingency advisory for Maharashtra state for aberrant weather:

There has been a delay in the onset of rainfall, and further, a long dry spell occurred during the first fortnight of July. If a dry spell of more than 15 days results in the appearance of pencil-width cracks in the soil, provide life-saving irrigation to the newly sown crop (< 15 days old) using sprinklers, perforated pipes, drip irrigation, or alternate furrow irrigation, depending on the availability of irrigation facilities. A prolonged dry spell may result in temporary hot and dry conditions. Such conditions are highly conducive to the incidence and development of dry root rot disease at the seedling stage.

At the initial appearance of disease symptoms, apply *Trichoderma* spp. (*T. harzianum* or *T. viride*) 1% WP @ 5 g per litre of water around the base of the affected plants as well as the surrounding healthy plants. Or prepare a solution of Carbendazim 50 WP at 2 g per litre of water and drench the soil around the base of the affected plants.

TELANGANA		Actual Rainfall in last week(mm) July							Predicted Rainfall in next week (mm) July				
		07	08	09	10	11	12	13	15	16	17	18	19
	Adilabad								0	0.1	0.7	1.7	1.3
	Warangal								0	0	1.8	1.9	2.9
	Khammam								0	0	2	3	2.9
	Karimnagar	0	0	0	1	0	0	0	0	0	1.8	1.7	2.6
	Mahabubnagar								0	0	0.3	3.2	1
Amount of rainfall & colour Code		0.1 to 2.4 mm		2.5 to 15.5 mm		15.6 to 64.4 mm			64.5 to 115.5 mm			115.6 to 204.4	
Rainfall category		Very light rainfall		Light rainfall		Moderate rainfall			Heavy rainfall			Very heavy rainfall	

Crop Condition:

At Warangal, field preparation has been completed. Sowing is yet to start.

At Adilabad, almost 100 per cent of the cotton area in the district has been sown after receipt of monsoon rains. Sprayed pre-emergence herbicide in the sown fields. .

Advisory:

At Warangal, farmers are advised to go for crop rotation for every 2 to 3 seasons. Take up deep summer ploughing to destroy pest life stages and weed residues in the soil. Soil incorporation of FYM should be done. Select medium duration hybrids. To manage soil borne diseases, at the time of sowing, treat the seed with fungicide Carboxin 37.5% + Thiram 37.5% DS @ 3.5 g per kg of seeds (root rot and bacterial blight disease) or Fluxapyroxad (333g/lit FS) @ 1.5 ml/kg@1.5ml/kg seed or Tetraconazole 11.6% w/w (12.5% w/v) SL @ 1.5 ml/kg of seeds for seed-borne borne diseases (for root rot disease). as well as biofertilizers, *viz.,* Azotobacter, PSB and KMB @ 10 ml per kg seed or *Trichoderma harzanium* 2.5 kg/ha enriched with 100 kg FYM. Sow the crop after receipt of at least 80-100 mm rainfall to avoid germination-related problems. Select the variety/hybrid based on soil suitability. Use pendimethalin 38.7% CS @ 700 ml/acre in 200 litres of water as a pre-emergence weedicide to control weeds in the early stage of the crop. Field and field bunds should be weed free from weeds and *Parthenium* weed should be eradicated in and around the fields to minimize the thrips population build-up and TSV incidence.

At Adilabad, farmers are advised to keep fields and field bunds free from weeds, and *Parthenium* weed should be eradicated in and around the fields to minimize the thrips population build-up and TSV incidence. Spray post emergence herbicide Quizalofop ethyl @ 2 ml or Propaquizofop @ 1.25 ml / lit for grassy weeds, Pyrithiobac sodium @ 1.25 ml/lit for broad leaves in cotton to control their spread.

ANDHRA PRADESH		Actual Rainfall in last week(mm)							Predicted Rainfall in next week (mm)				
		July							July				
		07	08	09	10	11	12	13	15	16	17	18	19
	Guntur								0	0	0.3	2.5	2.5
	Prakasam								0	0	0.2	2.8	1
Amount of rainfall & colour Code		0.1 to 2.4 mm		2.5 to 15.5 mm		15.6 to 64.4 mm			64.5 to 115.5 mm			115.6 to 204.4	
Rainfall category		Very light rainfall		Light rainfall		Moderate rainfall			Heavy rainfall			Very heavy rainfall	

Crop Condition:

At Guntur, sowings will be taken up only after the receipt of sufficient monsoon rains.

At Nandyal, nearly 65 per cent of the cotton area in Kurnool district was sown during first fortnight of June under low soil moisture conditions. In the majority of areas, crops are exhibiting wilting and moisture stress symptoms. The crop is at 10-40 days old at seedling to vegetative stage. Sucking pests were above ETL in certain locations. Inter-cultivation was taken up during the reporting period.

Advisory

Farmers are advised to take up deep summer ploughing to kill pupae of pink bollworm and seeds of weeds. Soil incorporation of FYM should be done. Select medium duration hybrids. To manage soil borne diseases, at the time of sowing, treat the seed with fungicide Carboxin 37.5% + Thiram 37.5% DS @ 3.5 g per kg of seeds (root rot and bacterial blight disease) or Fluxapyroxad (333g/lit FS) @ 1.5 ml/kg@1.5ml/kg seed or Tetraconazole 11.6% w/w (12.5% w/v) SL @ 1.5 ml/kg of seeds for seed-borne borne diseases (for root rot disease). as well as biofertilizers, *viz.,* Azotobacter, PSB and KMB @ 10 ml per kg seed or *Trichoderma harzanium* 2.5kg/ha enriched with 100 kg FYM. Sow the crop after receipt of at least 80-100 mm rainfall to avoid germination related problems. Use Pendimethalin 38.7 % CS @ 700 ml/acre in 200 litres of water as pre-emergence weedicide to control weeds in the early stage of crop.

KARNATAKA		Actual Rainfall in last week(mm)							Predicted Rainfall in next week (mm)				
		July							July				
		07	08	09	10	11	12	13	15	16	17	18	19
	Dharwad	31	10.8	11.2	0	0	0	0	0	1.1	2.3	5	2
	Haveri								0	0.4	0.7	5	3.4
	Mysore								0	0	0.4	3.6	1.8
Amount of rainfall & colour Code		0.1 to 2.4 mm		2.5 to 15.5 mm		15.6 to 64.4 mm			64.5 to 115.5 mm			115.6 to 204.4	
Rainfall category		Very light rainfall		Light rainfall		Moderate rainfall			Heavy rainfall			Very heavy rainfall	

Crop Condition:

At Dharwad, the crop is 15 to 20 days from germination to seedling stage. Sporadic rainfall received in surrounding locations. Hand weeding and inter cultivation for weed management is in progress. Take up sowing of okra for every 20 rows of cotton for Shoot weevil pest management at the time of sowing of cotton. No incidence of weeds, pests, or diseases.

At Raichur, nearly 20-30 per cent of first sown crop has already withered out and nearly 20 % area has been resown with cotton again. Designated irrigated lands are waiting to receive irrigation water and there is less possibility of release of canal water for irrigation. Early sown crops are 25-30 days old during the reporting time. Sowing operation has been taken up in some irrigated patches and rainfed areas. Remaining farmers have been advised to take up land preparations during this week. Pre-emergent application of Pendimethalin @ 3.5 ml per litre of water has been advised to the farmers who have taken up sowings.

At Chamarajanagar, the crop is 51 to 66 days at the square formation to the flowering stage. Top dressing with Urea fertilizer, weeding, inter-cultivation, and earthing up work operations are going on. Infestations of leafhoppers and aphids were noticed.

Advisory:

At Dharwad, farmers are advised to sow Okra for every 20 rows of cotton for shoot weevil management.

At Raichur, farmers are advised to take up weeding and inter-cultivation operations on a priority. Give post-emergence herbicide spray with Pyrithiobac Sodium @ 1.5 ml per litre of water when the crop is at 25-30 days duration. The early sown

crop is recommended with the first top dressing with Urea and MOP. Spray the recommended dose of Flonicamid 50% WG (4 g/ 10 litres of water) against sucking pest populations exceeds ETL.

At Chamarajanagar, farmers are advised to install yellow sticky traps @ 8-12 per acre for monitoring of sucking pests. Spray Imidacloprid 17.8 SL @ 0.3 ml /lit of water to control sucking pests.

TAMIL NADU		Actual Rainfall in last week(mm)							Predicted Rainfall in next week (mm)				
		July							July				
		07	08	09	10	11	12	13	15	16	17	18	19
	Perambalur								0	0	0	0.5	0.7
	Salem								0	0	0	1.3	0.6
	Trichy								0	0	0	0.5	0.2
	Virudhunagar								0	0	0	0.2	0.6
Amount of rainfall & colour Code		0.1 to 2.4 mm		2.5 to 15.5 mm		15.6 to 64.4 mm			64.5 to 115.5 mm			115.6 to 204.4 mm	
Rainfall category		Very light rainfall		Light rainfall		Moderate rainfall			Heavy rainfall			Very heavy rainfall	

Crop Condition:

Field preparation is in progress.

Advisory

Farmers are advised to take up deep summer ploughing to kill pupae of pink bollworm and other weed seeds. Soil incorporation of FYM should be done. Select medium-duration hybrids. To manage soil-borne diseases, at the time of sowing, treat the seeds with Carboxin 37.5% + Thiram 37.5% DS) @3.5 g per kg of seeds or soil application of *Trichoderma harzanium/ T. viride* 2.5kg/ha enriched with 100 kg FYM. Sow the crop after receipt of at least 80-100 mm rainfall to avoid germination-related problems.

C. Post-season and pre-sowing package of practices

1. Clean up fields of residual stalks and partially opened bolls from the previous crop season. Do not stack the uprooted cotton stalks on field bunds. At the end of crop season, the pink bollworm larvae of last generation enter hibernation in crop residues like infested bolls. Therefore, such infested residues should be promptly destroyed in order to break the life cycle of pink bollworm. Residue destruction will also help to reduce the inoculums and infection of new season's cotton crop by diseases like bacterial leaf blight, root rot and fungal leaf spots.
2. Install at least 10 pheromone traps for pink bollworm each at 20 m distance in the premises of market yards and ginning mills to trap post season moths or suicidal emergence if any. Change the lures in pheromone traps timely. Also kill the larvae that come out of damaged seeds. This will help to check the spread of infestation of pink bollworm from ginning or market yard premises to nearby fields.
3. Avoid pre-monsoon sowing of cotton crops. Early sown crops bear reproductive structures like squares and flowers early. The pink bollworm moths emerging from dormant populations of the previous season lay eggs on these squares and flowers; thus, an early-sown crop supports completion of the new season's first generation of pink bollworm. If not controlled timely, next generations of this population further spreads onto the timely sown cotton crop with onset of squares, flowers and bolls.
4. Deep summer ploughing helps to expose and kill the dormant larvae and pupae hidden in the soil due to the scorching heat of sun and predation by birds. in April-May. Also, the birds following ploughed fields predate these life stages of insects. This helps in minimising the incidence of insects like pink bollworm, leaf eating caterpillars, and soil-borne diseases like wilt, root rot and nematodes on the coming season's cotton crop.
5. Crop rotation to be followed in the fields that were heavily infested with pink bollworm during last season to break the life cycle of pink bollworm. Cotton is the only host of pink bollworm, therefore crop rotation helps to break the life cycle of this pest. Crop rotation is very effective in checking the infection of soil borne diseases and nematodes in disease prone fields.
6. Grow sucking pest and disease-tolerant, short-duration and early maturing varieties/hybrids/cultivars of cotton. This helps in avoiding unwanted spraying of pesticides to control sucking pests and diseases during the early crop growth stage. Pink bollworm infestation starts from mid-season and increases steadily towards the late season. Therefore, short duration and early-maturing varieties help to escape pink bollworm infestation in late season.
7. Sowing of cotton crops should be done in the month of June, only after receipt of 80-100 mm of monsoon rainfall. For ensuring proper germination and crop stand to withstand the prolonged dry periods during early seedling stage, there should be optimum soil moisture. This also helps to avoid re-sowing due to a prolonged dry spell of rainfall. Timely sowing in June helps to avoid early infestations of pink bollworm.
8. Increased awareness should be created among the cotton farmers regarding the implementation of an integrated pest management (IPM) strategy for the management of pink bollworm. The input dealers may also be advised to inform the farmers not to adopt pre-monsoon sowing. This will help to spread the right message to farmers more effectively

The detailed information regarding cotton production technology, e.g., selection of soil, varieties, fertilizer application, sowing methods, irrigation systems, management of weeds, insect pests and diseases, etc. can be availed from an Android-based **CICR Cotton App** developed by ICAR-CICR, Nagpur. The app can be downloaded free of cost from Google play store. Additionally, the crop growth stage-specific and weather-based weekly advisory are uploaded on the website of ICAR-CICR (https://cicr.org.in/resource-weekly-advisory) also to be consulted for the benefit of farmers.

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